# MLIP + Distill: A Shared Bias Direction in Cubic-Metal Elastic Constants

## A 3×3×3 Reference Benchmark and the Oracle Ceiling of a Post-Hoc Correction

**Lupine Project**
*Correspondence: alex@lupinesci.com*
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> **Status: Working draft, not peer-reviewed. Correction 2026-11-11.** Earlier wording in the abstract, discussion and conclusion said every model improves and that a family-level correction "repairs every member". The leave-one-out result below uses the held-out reference to set how far to correct, so 17.84 → 10.36 GPa is an **oracle ceiling, not a deployable correction**. The best no-target estimator later tested on this benchmark (consensus magnitude, 14.41 GPa) harmed 50 of 128 cases (ProjectionLaw_Round2.md). The registered Round-2 functional-clustering predictions (H1, H2a) failed, and r2SCAN targets here are approximate (scalar bulk-modulus shifts of PBE). Status: conjecture ledger.

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## Abstract

We report a 128-case 3×3×3 elastic-constant reference matrix for four MatPES foundation machine-learned interatomic potential (MLIP) labels on 16 cubic metals, against PBE and approximate r2SCAN targets (scalar bulk-modulus shifts of PBE). The matrix costs less than one CPU core-hour; raw predictions have a mean $C_{ij}$ MAE of 17.8 GPa. One bias direction per functional, fitted leave-one-out, transfers out of sample. When the held-out reference sets how far to correct along that direction, the mean MAE falls to **10.4 GPa** (PBE 15.0 → 9.4 GPa; r2SCAN 20.7 → 11.3 GPa). Because the held-out target sets the coefficient, this is an **oracle ceiling, not a deployable correction**, and it cannot increase a held-out residual by construction. A correction that works without seeing the answer is still open. Error is strongly stratified by chemistry: alkaline-earth and noble FCC metals are already accurate (Ca 2.9 GPa), while magnetic and refractory BCC metals remain the frontier (Cr 43.5 GPa).

**Certification status:** The corrected $C_{ij}$ MAEs below are empirical/oracle aggregates, not licensed correction guarantees. They are **uncertified** unless C11, C12, and C44 each independently satisfy a valid per-component license for the held-out target. Bulk/shear moduli, $C'$, Cauchy pressure, anisotropy, and other derived quantities are also **uncertified**: componentwise licenses do not compose through differences, mixed directions, or general products.

**Keywords:** machine-learned interatomic potentials, elastic constants, correction operator, MatPES, benchmark, supercell convergence, error geometry

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## 1. Introduction

Materials discovery pipelines rely on elastic constants as an early filter. The standard way to control error is to pay for ensembles of independent models or for large supercells. Both multiply cost. Foundation MLIPs trained on DFT corpora promise a cheaper path, but their model-form error is material-dependent and often systematic: the same training functional imparts the same stiffness bias to every architecture that learns from it.

The Projection Law formalizes this observation [1]. A model family is a projection operator; fitting drives every member toward the nearest point of the family's reachable set; the shared residual is a fingerprint of the binding constraint. The practical corollary is that one correction direction per (constraint, observable) can repair every model in the family at once — provided the direction is identified and validated out-of-sample.

Here we test that corollary on the lowest-risk, highest-throughput corner of materials space: cubic elastic constants of 16 elemental metals. We establish a complete 3×3×3 reference matrix for four MatPES foundation MLIPs (CHGNet, M3GNet, QET, TensorNet) under PBE and approximate r2SCAN targets. We then extract a single one-dimensional bias vector per functional from the residual cloud and apply it in leave-one-out cross-validation. With the held-out target setting the projection coefficient, the operator lowers the mean MAE for every model and functional combination; that is an oracle ceiling, not a deployable result. The remaining uncorrected error is concentrated where the shared-bias assumption breaks down: magnetic and refractory BCC transition metals. The benchmark and the operator are the two products; together they give a fast, diagnostic workflow for cubic-metal elasticity.

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## 2. Methods

### 2.1 Benchmark set

The target set is 16 cubic elemental metals: Ag, Al, Au, Ca, Cr, Cu, Fe, Mo, Nb, Ni, Pd, Pt, Sr, Ta, V, and W. For each element we compute the three independent elastic constants from a conventional cubic cell relaxed and then expanded to a 3×3×3 supercell (108 atoms for FCC, 54 atoms for BCC).

The MLIPs are the MatPES 2025.2 foundation models loaded through `matcalc` [2]:

- CHGNet [3]
- M3GNet [4]
- QET
- TensorNet

QET and TensorNet are closely related TensorNet-family models. In earlier Lupine work using a non-PES loader the two labels resolved to a common checkpoint and were treated as aliases [5]. In the PES-labeled 2025.2 release used here they return different predictions; we report them as distinct model objects while noting that the architectural comparison is not clean.

Each model is evaluated against two targets:

- **PBE:** 0 K elastic tensors from de Jong *et al.* 2015 [6], with the Ag tensor from Pandit & Bongiorno 2023 [7] and a PW91-GGA fallback for Au from Wang & Li 2008 [8].
- **r2SCAN:** PBE tensors scaled by a scalar bulk-modulus ratio from Liu *et al.* 2024 [9]. Al, Ca, and Sr retain a shift factor of 1.0 because no r2SCAN bulk modulus was recovered. The r2SCAN comparison is a sensitivity check, not a ground-truth claim.

### 2.2 Computational workflow

The workflow uses `matcalc` with a standardized stress/strain elasticity calculator:

1. Build the conventional cubic cell at the starting lattice constants in `lupine/data/layer2_benchmark_task.py`.
2. Expand to a 3×3×3 supercell.
3. Relax cell and positions with `RelaxCalc` (fmax = 0.005 eV/Å).
4. Compute the elastic tensor with `ElasticityCalc` (fmax = 0.005 eV/Å, GPa units).
5. Extract $C_{11}$, $C_{12}$, and $C_{44}$.

Wall-clock runtime is recorded. CPU-equivalent core-hours are `runtime_seconds / 3600`, excluding one-time model downloads. The 128-case matrix was executed as a Cloud Run job array in GCP project `witching-606c6`, region `us-central1`, container image `us-central1-docker.pkg.dev/witching-606c6/lupine-layer2/runner:v1`. Outputs were uploaded to `gs://lupine-benchmark-witching-606c6/layer2_3x3x3/` and aggregated with `lupine/data/aggregate_layer2.py`.

### 2.3 The Lupine correction operator

For a given functional, stack the raw predictions and reference targets as 3-vectors of ($C_{11}$, $C_{12}$, $C_{44}$). The residual matrix is `R = target − pred`. The Lupine correction direction is the first principal component of `R`, normalized to a unit vector **b**. For any residual **r**, the best one-dimensional correction is the projection of **r** onto **b**:

$\alpha = (\mathbf{r} \cdot \mathbf{b}) / (\mathbf{b} \cdot \mathbf{b}) = \mathbf{r} \cdot \mathbf{b}$,

corrected = pred + $\alpha$ **b**.

By construction the projection cannot increase the Euclidean norm of the residual, so the operator satisfies a no-harm guarantee on the rows used to define it.

### 2.4 Leave-one-out validation

To test whether the bias *direction* transfers to unseen rows, we use leave-one-out cross-validation. For each of the 128 cases, the bias vector **b** is extracted from the residuals of the other 63 cases of the same functional. The held-out residual is then projected onto **b** to obtain the corrected prediction. This measures the out-of-sample transferability of the direction; it still uses the held-out target to set the projection magnitude, so it is an oracle-style ceiling for a no-target operator. The no-harm property is checked on every held-out row.

Uncertainty is quantified with percentile bootstrap confidence intervals (10,000 resamples with replacement over cases).

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## 3. Results

### 3.1 Raw benchmark

Table 1 reports the raw mean $C_{ij}$ MAE by model and functional. QET has the lowest raw MAE; CHGNet is the highest.

**Table 1 — Raw mean $C_{ij}$ MAE (GPa).**

| Model | PBE | r2SCAN | Overall |
|---:|---:|---:|---:|
| CHGNet | 17.90 | 27.94 | 22.92 |
| M3GNet | 14.13 | 20.71 | 17.42 |
| TensorNet | 14.61 | 18.54 | 16.58 |
| QET | 13.41 | 15.46 | 14.44 |
| **All models** | **15.01** | **20.66** | **17.84** |

PBE-trained models outperform r2SCAN-trained models across all four labels, with a mean functional gap of 5.7 GPa.

### 3.2 LOO-corrected benchmark

Table 2 reports the LOO-corrected MAE (oracle ceiling: the held-out target sets the projection coefficient; see §4.2). Under that oracle, the correction improves every model on both functionals. The overall mean MAE falls from 17.84 GPa to 10.36 GPa; the 95% bootstrap CI for the corrected mean is [8.9, 12.0]. No held-out row has a larger Euclidean residual after correction.

**Table 2 — Raw versus LOO-corrected mean $C_{ij}$ MAE (GPa).**

| Model | PBE raw | PBE LOO-corr. | r2SCAN raw | r2SCAN LOO-corr. | Overall raw | Overall LOO-corr. |
|---|---|---|---|---|---|---|
| CHGNet | 17.90 | 11.01 | 27.94 | 13.57 | 22.92 | 12.29 |
| M3GNet | 14.13 | 8.37 | 20.71 | 11.82 | 17.42 | 10.09 |
| QET | 13.41 | 9.22 | 15.46 | 8.69 | 14.44 | 8.95 |
| TensorNet | 14.61 | 8.97 | 18.54 | 11.22 | 16.58 | 10.09 |
| **All models** | **15.01** | **9.39** | **20.66** | **11.32** | **17.84** | **10.36** |

The direction transferability is the central result: a single bias vector fitted on 63 cases transfers to the 64th. The roughly 40% error reduction is an oracle ceiling, because the held-out target sets how far to move along that vector.

### 3.3 Per-element error landscape

Table 3 ranks elements by mean raw MAE across all models and functionals. The easiest systems are FCC alkaline-earth and noble metals; the hardest are BCC transition metals.

**Table 3 — Per-element mean raw $C_{ij}$ MAE (GPa).**

| Rank | Element | Mean MAE | Best model (functional) | Best MAE |
|---|---|---|---|---|
| 1 | Ca | 2.87 | CHGNet (r2SCAN) | 1.47 |
| 2 | Sr | 3.98 | CHGNet (PBE) | 1.93 |
| 3 | Ag | 7.30 | M3GNet (PBE) | 3.58 |
| 4 | Ni | 11.23 | M3GNet (PBE) | 3.43 |
| 5 | Pd | 12.20 | TensorNet (PBE) | 6.55 |
| 6 | Cu | 14.35 | TensorNet (PBE) | 9.73 |
| 7 | Al | 15.51 | M3GNet (PBE) | 7.35 |
| 8 | Au | 16.12 | QET (r2SCAN) | 4.71 |
| 9 | Ta | 17.00 | QET (PBE) | 8.64 |
| 10 | Mo | 19.94 | M3GNet (r2SCAN) | 8.82 |
| 11 | W | 20.79 | TensorNet (r2SCAN) | 7.84 |
| 12 | Pt | 23.07 | M3GNet (r2SCAN) | 6.27 |
| 13 | Fe | 23.29 | QET (PBE) | 8.86 |
| 14 | Nb | 26.92 | TensorNet (PBE) | 21.92 |

| Rank | Element | Mean MAE | Best model (functional) | Best MAE |
|---|---|---|---|---|
| 15 | V | 27.38 | TensorNet (PBE) | 13.97 |
| 16 | Cr | 43.47 | QET (PBE) | 5.72 |

The correction removes most of the bulk-stiffness error, leaving chemistry-specific errors concentrated in magnetic and refractory BCC metals.

### 3.4 Cost

The full 128-case 3×3×3 matrix costs approximately **0.82 CPU core-hours** in cache-warm, single-process CPU time.

**Table 4 — Total CPU core-hours by model.**

| Model | Total core-hours | Mean seconds / case |
|---|---|---|
| M3GNet | 0.075 | 8.4 |
| CHGNet | 0.431 | 48.5 |
| QET | 0.156 | 17.6 |
| TensorNet | 0.158 | 17.8 |

The correction itself is a deterministic vector projection and adds no inference cost.

### 3.5 Systematic signatures

Mean signed errors reveal model-specific signatures (Table 5). Values are averages over both functionals.

**Table 5 — Mean signed errors (GPa) and bulk/shear moduli biases.**

| Model | $\langle \Delta C_{11} \rangle$ | $\langle \Delta C_{12} \rangle$ | $\langle \Delta C_{44} \rangle$ | $\langle \Delta B \rangle$ | $\langle \Delta G \rangle$ |
|---|---|---|---|---|---|
| CHGNet | −23.28 | −1.98 | +1.74 | −9.08 | −0.52 |
| M3GNet | +4.49 | −9.35 | +8.36 | −4.80 | +7.52 |
| QET | +15.60 | −4.59 | +4.71 | +2.14 | +5.91 |
| TensorNet | −9.89 | −10.91 | +1.62 | −10.57 | +1.29 |

These signatures are exactly what a family-level correction should remove: a shared stiffness bias that persists across elements and functionals.

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## 4. Discussion

### 4.1 The operator works because the bias is shared

The LOO result tests the direction, not a deployable correction. The correction direction is never fit to the row it is scoring, and with the held-out target setting the coefficient it improves 128 of 128 cases in MAE terms; it cannot increase the Euclidean residual norm, by construction. The size of the improvement indicates that the MLIP errors share a low-dimensional direction on this benchmark. The first principal component lies predominantly in the $C_{11}$–$C_{12}$ bulk plane and is similar for PBE and r2SCAN, which is why one vector per functional suffices.

### 4.2 From oracle to deployable operator

The LOO operator uses the held-out target to set the projection magnitude. A deployable operator must set that magnitude without the target. The program's earlier operator-failure diagnosis identified two practical candidates [10]:

- **`scalar-bulk`** — use the scalar PBE-to-r2SCAN bulk-modulus shift as a proxy for the residual magnitude. On a 16-element TensorNet/PBE benchmark it achieved 14.13 GPa against the approximate r2SCAN target only; it did not beat the raw model on the PBE target.
- **`feedback-projection`** — fit the projection coefficient on a calibration set and apply the direction to new cases. It improved over `scalar-bulk` in 1×1×1 tests.

The 3×3×3 LOO ceiling (10.4 GPa) bounds how much these no-target operators can improve. These numbers are not directly comparable: 10.4 GPa is the four-model oracle ceiling on both targets, while `scalar-bulk`'s 14.13 GPa is one model (TensorNet/PBE, 1×1×1) against the approximate r2SCAN target only. The best no-target estimator tested on the full 128-case matrix (consensus) reached 14.41 GPa and harmed 50 of 128 cases. Closing the gap to the ceiling without seeing the reference is still open.

### 4.3 The remaining frontier

Even after the LOO correction, the mean error is ~10 GPa. The residual is not random: it is concentrated in magnetic and refractory BCC metals where the shared bulk-stiffness assumption fails. Cr, Fe, Mo, V, and Nb retain large errors because their errors are not aligned with the global bulk bias. These are the cases a class-aware operator would need to partition out. For alkaline-earth and noble FCC metals, the correction already brings most predictions within the uncertainty of the reference data.

### 4.4 Relation to the Projection Law

This benchmark was the Layer-2 test bed for the Projection Law hypothesis [1]. The pre-registered hypotheses H1–H4 have since been evaluated on the 128-case matrix: H1 and H2a failed (functional-clustering effect −0.14, permutation $p = 0.13$), H2b's cosine was 0.53 against a registered 0.8, and H4's conformal coverage was 86.7% against a registered 90%; H3 is pending (see the correction note at the top). The LOO result shows that a bias direction transfers; it does not show that a family-level correction repairs every member.

## 5. Limitations

- **Oracle magnitude.** The LOO correction uses the held-out target to set the projection coefficient. It proves direction transferability but is not yet a no-target operator.
- **Approximate r2SCAN targets.** r2SCAN tensors are scalar bulk-modulus shifts of PBE tensors, so the r2SCAN comparison is a sensitivity check.
- **Small model count.** Four model labels are evaluated, but QET and TensorNet are TensorNet-family variants; the clean architectural comparison is between CHGNet, M3GNet, and TensorNet.
- **No spin polarization.** Magnetic elements were run non-spin-polarized, which may inflate their errors.
- **Cubic elements only.** Results should not be extrapolated to lower-symmetry crystals, defects, alloys, or finite-temperature properties.
- **Single configuration per case.** Run-to-run variance has not been quantified for the full grid.

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## 6. Conclusion

We present a 3×3×3 elastic-constant reference for 16 cubic metals and four MatPES foundation MLIP labels (three architectures, since QET and TensorNet resolve to the same checkpoint), against PBE and approximate r2SCAN targets. One bias direction per functional transfers out of sample. When the held-out reference sets how far to correct, the benchmark mean MAE falls from 17.8 GPa to 10.4 GPa: an oracle ceiling, not a deployable correction. A correction that works without seeing the answer is still open; the best no-target estimator tested so far improved the mean but harmed 50 of 128 cases. The remaining error is concentrated in magnetic and refractory BCC metals. The useful product today is the cheap reference matrix and the error map, not a corrected calculator.

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## Data availability

- Raw outputs: `gs://lupine-benchmark-witching-606c6/layer2_3x3x3/*.json` (128 files)
- Summary JSON: `lupine/data/benchmark_layer2_3x3x3_summary.json`
- Source repository: `https://github.com/alexwelcing/lupine`
- Pre-registration and companion results: `lupine-rhizo/docs/projection-law-round2-preregistration.md`, `lupine-rhizo/docs/projection-law-round2-results.md`
- Operator-failure diagnosis: `lupine-rhizo/mlip-elastic-benchmark/operator-failure-diagnosis-2026-06-27.md`

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